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RESEARCH ARTICLE

Technoeconomic Assessment of Electric Vehicle Battery Disassembly—Challenges and Opportunities From a Robotics Perspective

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ABSTRACT The rapid shift towards electric vehicles (EVs) demands effective end-of-life strategies for lithium-ion batteries (LIBs), necessitating examining recycling methodologies, particularly the disassembly process. This study presents a technoeconomic analysis of EV battery disassembly, focusing on incorporating robotics to address challenges and capitalize on opportunities. Based on the case study of the Mitsubishi Outlander PHEV battery pack, we identify the most labor and cost-intensive components and introduce a structured approach to evaluate automating disassembly tasks. Classifying tasks based on their automation potential, we find 57% of pack-to-module (P2M) tasks readily automatable, with an additional 24% requiring minimal human intervention. Incorporating this into an analysis of task feasibility, level of automation and robot time efficiency, we establish a roadmap towards robotizing the disassembly process based on human-robot collaboration, demonstrating a 12.85% increase in net annual revenue and 43.75% reduction in disassembly cost with current level of task feasibility. Our findings indicate a long-term roadmap towards increasing level of automation, while surmounting feasibility of complex disassembly tasks remains an unsolved challenge.

INDEX TERMS Circular economy, electric vehicles, lithium-ion batteries, recycling, robotic disassembly.

I. INTRODUCTION

The increasing electrification of the automotive industry and the spread of electric vehicles (EVs) represent a shift in the transportation landscape. Demand for high-capacity lithium-ion batteries (LIBs) is expected to increase year-on-year [1], which introduces medium- and long-term political, economic and sustainability challenges, driven by concerns over medium-term material availability and limited battery life cycle [2]. In particular, the progressing life cycle of EV batteries [3] represents a challenge for end-of-life disposal and recovery of critical materials. Reusing and recycling batteries at the end-of-life stage is expected to play a

key role in recovering these materials and minimizing the environmental impact associated with their disposal. Whereas disassembly forms an essential pillar of existing recycling processes and a key part of battery second life application, *cost-effective* disassembly remains a complex problem due to labor costs, variety in battery designs, and current volumes of battery waste.

Previous work has conducted technical and economic analyses of specific aspects of the battery disassembly process. For example, Yu et al. [4] emphasize the importance of improving LIB recycling processes in the context of the growing EV industry, and provide insights into the current challenges, including recycling process challenges, legislative challenges, and costs, while also providing possible strategies for more efficient and sustainable recycling

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practices. Zorn et al. [5] offers a thorough approach to addressing the challenges of disassembling and recycling lithium-ion battery packs, pointing out the need for automation and the latent benefits of using computer vision, labeling, and material characterization techniques. Lander et al. [6] explored the economic impact of battery pack design on the disassembly costs and the benefits of automation in terms of cost and time efficiency across six different battery packs. The study also identifies common challenges in battery pack disassembly, primarily due to the need for more standardization across different manufacturers. While it focuses on analyzing cost-intensive design features, it lacks an exploration of the feasibility of automation for specific tasks and a detailed breakdown of the disassembly process. Baazouzi et al. [7] explored the optimization of an EV battery disassembly process to reduce costs, concluding that disassembly strategies need to consist of three decisions about the optimal disassembly sequence, disassembly depth, and circular economy strategy for each component in the battery pack. Remanufacturing of EV batteries has also been explored from an economic and environmental perspective for second-life applications [8]. Closest to this work, Villagrossi and Dinon [9] investigated the potential for automation based on a battery disassembly case study of the Stellantis 500e MY 2022 battery pack, studying the disassembly steps, principal disassembly tasks and their automation potential. In this case, the authors proposed a robotic disassembly cell concept, however, did not consider disassembly task completion times or effect of automation on disassembly costs and revenues.

Also contributing to the literature, Li et al. [10] explored the possibilities of utilizing robotic disassembly to increase the purity of high-value waste from EV batteries. The authors explore different strategies for automated disassembly, albeit with a focus on the recovery of electronic components such as printed circuit boards (PCBs), observing the lack of recycling automation, making the process more time-consuming than an automated process. Blankemeyera et al. [11] assessed tasks required for EV battery disassembly, focusing on handling and types of connections between components, with the authors proposing a concept for pack-to-module (P2M) disassembly cell but not considering module-to-cell (M2C) disassembly, nor the potential cost of disassembly calculations. Other works have focused on the M2C disassembly process, such as that by Gerlitz et al. [12], which addressed tasks required for M2C disassembly, focusing on developing an agile, automated disassembly system and exploring both the potential benefits and the challenges, such as variety of battery module designs and process-related complexities of dealing with LIBs. Further work in M2C disassembly was undertaken in [13] which explored the potential of robotic systems in automating the disassembly process of EV battery modules, highlighting benefits in terms of safety and efficiency and suggesting the optimal collaboration between human technicians and robots. Schäfer et al. [14]

discussed challenges and potential solutions for automated disassembly and remanufacturing of EV battery modules, focusing on improving the disassembly process to facilitate remanufacturing. While these studies have concentrated on either pack-to-module or module-to-cell disassembly, there is a gap in the literature addressing both processes. Zhou et al. [15] discussed further challenges in recycling lithium-ion battery packs, analyzing benefits and limitations in automating the EV battery recycling process and required improvements in robot technologies for improving disassembly safety. Similarly, Meng et al. [16] provided an overview of the autonomous disassembly of EV batteries, exploring the automation potential for a range of disassembly tasks utilizing AI and ML for intelligent disassembly. Klohs et al. [17] provided an overview of the principal process and product-specific challenges of automated battery disassembly, including model variety, battery conditions uncertainty, difficulty with non-destructively detachable joints, and lack of data availability, based on interviews with domain experts. However, the study does not consider a concrete case of battery disassembly and does not consider the effect of automation or robotizing on productivity from a process perspective.

Some authors also provide their analysis of the disassembly process, each focusing on different parts of it and showcasing their benefits. Following the use of robotics in EV battery disassembly, as Zang and Wang [18] add, it would follow that labor cost would sharply decline, with also a substantial drop in processing time, depending on the complexity and variation of the configurations in the disassembly of battery packs. More important is the effectiveness of the human-machine collaboration that Wang et al. [19] signal, which promises to improve task completion times due to the integration of artificial intelligence algorithms for task sequencing optimization and identification of operational targets. Zhang et al. [20] document NeuroSymbolic computing in the use case of autonomous disassembly, noting its gains in increased disassembly precision and improvement of the process efficiency through dynamic task and motion planning, especially in unstructured environments with obstacles. This analysis, therefore, collectively points toward a persuasive case for the robotization of EV battery disassembly, showing the economic, safety, and efficiency benefits needed for scaling sustainable recycling operations.

This paper focuses on the *pack-to-module* and *pack-to-cell* disassembly process of the Mitsubishi Outlander PHEV (plug-in hybrid electric vehicle) battery pack, with an emphasis on automation potential for disassembly from a robotics perspective. The main contributions of this work are a taxonomic study of task feasibility based on a detailed breakdown of the pack-to-module and pack-to-cell disassembly process, an analysis of disassembly costs and revenues for robotizing with commercially available robot platforms and a roadmap towards implementation strategies

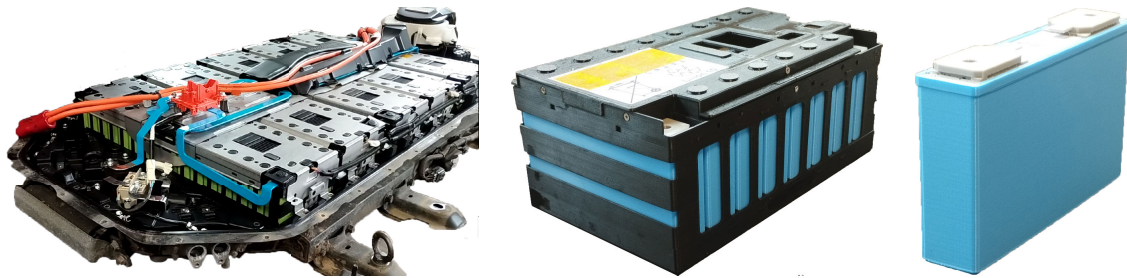


FIGURE 1. Overview of structure of Mitsubishi Outlander PHEV from battery pack (left, top case removed), module (centre), and prismatic cell (right).

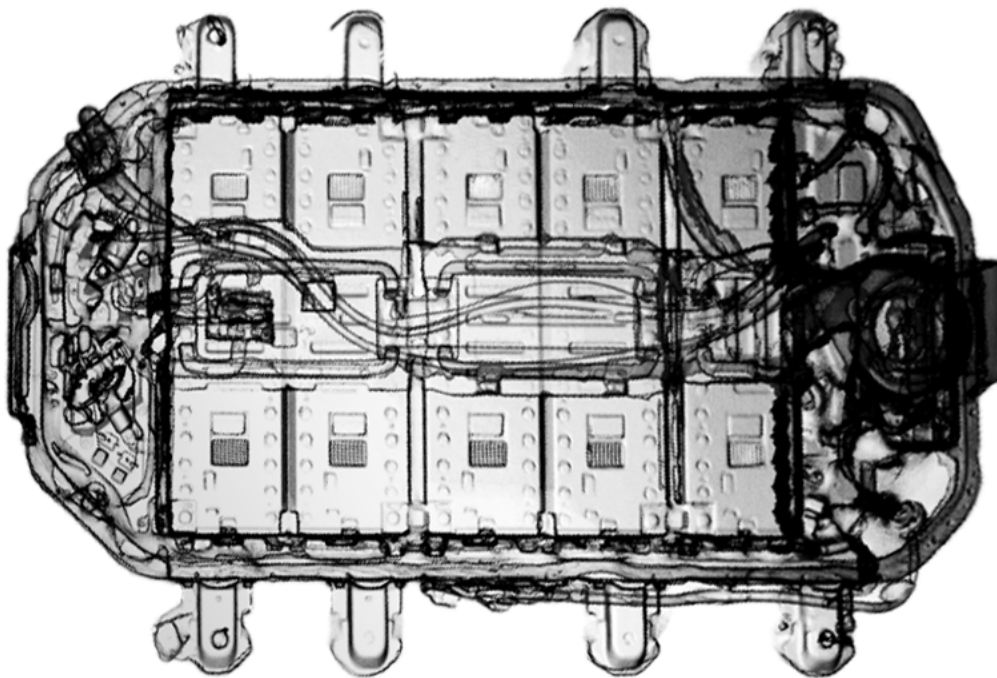


FIGURE 2. 3D scan of Mitsubishi Outlander PHEV battery pack, overhead view.

of robots for disassembly based on level of automation and robot task efficiency relative to human workers.

Section II provides an overview of the battery system, safety considerations, and analysis of the needed hierarchical flow for the automation process. Section III details the disassembly procedure and proposes a task categorization, separating disassembly tasks into fully automatable, semi-automatable, and manual processes, considering the full disassembly of the Mitsubishi Outlander PHEV. Section IV performs a detailed economic evaluation of robotizing the disassembly of battery packs. Section V incorporates an analysis of task feasibility on disassembly cost, proposing methods of adapting manual tasks to autonomous, based on previous robot disassembly experiments. Finally, Section VI concludes with the insights and proposes potential future work to extend current limitations in the disassembly process and use to further improve the circular economy of batteries.

II. BATTERY DISASSEMBLY PROCESS

A. TRACTION BATTERY STRUCTURE

The traction battery is a key component in electric vehicle design, and exists in three main structural forms: pack-module-cell, cell-to-pack, and cell-to-structure [21] hierarchies. The former design remains the most prevalent and is the focus of this study based on the case study of the Mitsubishi Outlander PHEV. As an overview, the Outlander PHEV battery pack is rated for 12KWh, with a modular structure comprising ten modules with eight prismatic format cells in an 8S configuration. Fig. 1-2 provides an outline of the internal components and arrangement of the pack. In the pack-module-cell hierarchy, individual battery cells are arranged into larger groups as modules, which provide protection to the cells and structural rigidity to the larger pack. Modules are arranged within the battery housing and interconnected through bus bars, which create high-voltage connections for power transmission between

modules. Harness connectors link the modules to the Battery Management System (BMS) and the Thermal Management System (TMS). The BMS manages key functions, such as monitoring the state of charge (SOC) and state of health (SOH), balancing cell charge, and protecting against overcharging and deep discharging. The TMS regulates thermal conditions, ensuring cell temperatures remain within operational limits.

Disassembly of the battery packs can involve dealing with these and further interconnected components, via a mixture of non-destructively detachable joints, including bolts, screws, clips, plugs and adhesives [6]. Traditionally, robots are optimized for carrying out repetitive, simple or precise manipulations but handle uncertain and unstructured environments poorly. In contrast, human operators are adept at dexterous manipulations yet are limited in precision, physical workload capacity and time efficiency for repetitive tasks. For example, housing or structural components are typically rigid objects affixed with threaded fasteners such as bolts in predetermined locations, whereas harness connectors are deformable objects with inherent position uncertainty, where removal of affixing plugs or clips requires complex manipulations. A variety of design features, hence imposes constraints on task allocation and feasibility of robotic disassembly. Therefore, understanding the function, theory, and integration of each component is necessary to incorporate robots into the disassembly process and ensure safe human-robot collaboration.

B. SAFETY AND ENVIRONMENTAL CONSIDERATIONS

Battery disassembly involves risks of electrical and chemical nature. There is a need to prioritize safety measures and environmental considerations to mitigate these risks and to ensure a secure working environment for technicians. Prior to disassembly, the battery undergoes extensive electrical testing and discharge, to make them safer to operate and to minimize the risk of unexpected electrical surges or faults during the disassembly. Personal Protective Equipment is an important part of the safety procedure; technicians must wear full-face visors and rubber gloves and use insulated tools during the disassembly. This equipment protects against chemical exposure and electrical shocks. Insulators are also placed on main harness connectors and module terminals to reduce the risk of accidental short circuits, sudden high current discharges, and thermal runaway. The current allocation of workload for battery disassembly is also motivated by safety considerations. For example, the battery represents a DC electrocution hazard which can lead to entrapment of a lone worker, hence, a risk mitigation strategy is to carry out disassembly with teams of multiple technicians. Post disassembly, the active battery components are subject to thermal deactivation, followed by shredding and sorting to separate the high-value components. The remainder of the waste from the disassembly process, consisting of low-value materials such as plastics and steel, is either disposed of or sent to a third-party recycling company for recovery.

C. PROCESS FLOW

We initially identify the sequence of disassembly tasks for *pack-to-module* and *pack-to-cell* disassembly for the Mitsubishi Outlander PHEV battery pack. To determine these steps, the full manual procedure, as carried out by a trained disassembly technician, was observed. During disassembly, the required tools, the number of operations per task, and the operation completion times were recorded. The disassembly steps documented were reported as closely as possible to the original process flow adopted during manual disassembly. However, in some cases, such as partially completed or interrupted tasks, we combine these split task segments into a single task if the latter task segments do not depend on later tasks to avoid unnecessary complexity in the documented disassembly sequence.

In the context of design features and safety considerations from Section II-A, II-B we assess the automation potential of each disassembly task based on the following considerations:

- a. *Identifiability of objects* - distinctive visual features, uniqueness, compatibility with sensors
- b. *Reachability of objects* - ease of access, visibility, few obstructions
- c. *Compatibility with tools* - graspable/flat surfaces, standard fasteners, low fastener variety
- d. *Manual dexterity* - simple manipulations vs. complex locking mechanisms, clips, plugs, etc.
- e. *Time consumption* - as compared with manual disassembly

Based on these considerations, we assign an estimated level of autonomy to each task. Fully autonomous (A) tasks can be accomplished with minimal or no user intervention, whereas semi-autonomous (SA) tasks require occasional manual interventions and conditioning for the specific battery model or are only partially autonomously completable. Fully manual (M) processes require a human operator to accomplish throughout or require dexterous manipulations that are highly challenging to achieve with a robot.

Broadly speaking, fully autonomous tasks benefit from the support of existing commercially available tools, simplicity, and task structure. These tasks are trivial to automate with a pre-programmed sequence of operations constructed for the specific battery model. However, these tasks can also be accomplished with general-purpose algorithms that can be employed across a range of battery designs without specialization on the specific battery model. The latter distinction is necessary due to the variety of battery designs due to the impracticality of reprogramming robots for each model. Semi-autonomous tasks share similarities with fully autonomous tasks, however, they are either challenging to automate with a pre-programmed sequence of operations (e.g. grasping deformable objects with pose uncertainty), challenging to accomplish with general-purpose algorithms (e.g. fasteners that are difficult to detect without design-specific guidance), or suffer from periodic failures that require manual intervention. Manual tasks are impossible

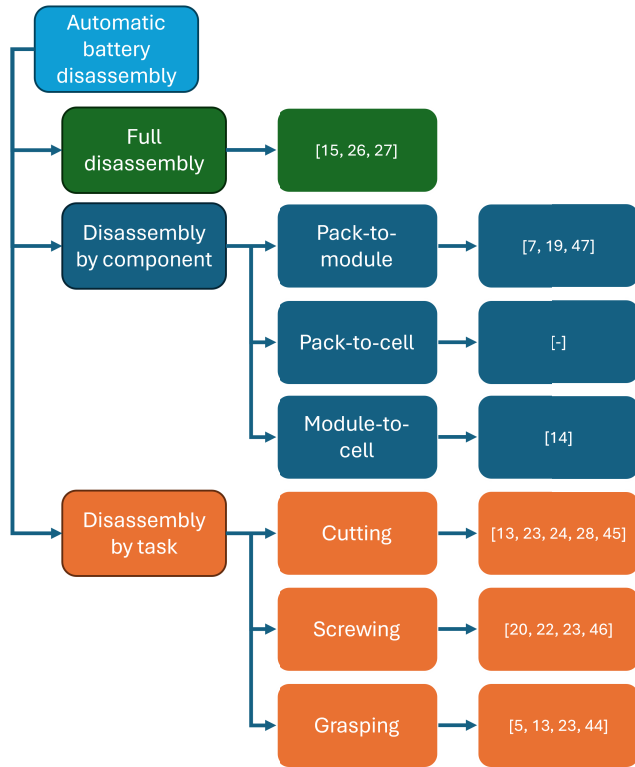


FIGURE 3. Studies on automatic battery disassembly, categorized by the type of disassembly considered.

to reliably accomplish with a pre-programmed sequence of tasks due to dexterous manipulations of components in combination with pose uncertainty and/or incompatibility with commercially available tools. While this does not entirely preclude automation, novel tools, developments in robot control, or scene understanding are required to attempt the task, with such methods typically being highly conditioned on the specific task. Literature studies included in this analysis on the automation of battery disassembly can be seen in Fig. 3, separated by their approach for the disassembly process, full disassembly, disassembly by component, and disassembly by task.

III. TECHNICAL EVALUATION

Based on the automation taxonomy and the recorded results according to Section II-C, the disassembly task sequence is reported in Table 1 and Table 2. In Table 1, semi-autonomous tasks are annotated with the principal limiting factor precluding full automation. Examples of such factors are obstruction of the components, such as bolts by other components, or the possibility of failures, such as grasping of complex objects with prehensile or suction-based grippers. The completion time for each task was measured directly from observing the manual process as per Section II-C.

A. PACK-TO-MODULE DISASSEMBLY

From the raw tasks recorded in Table 1–2, Fig. 4 shows a categorization and breakdown of task completion times by

required tools, indicating the type of task, and estimated level of autonomy according to the proposed taxonomy in Section II-C. For the pack-to-module case, shown in Fig. 4a–4b, an overwhelming majority ($\sim 76\%$) of disassembly time is spent on unbolting; a task which was predominantly categorized as fully autonomous (A) due to clear visibility and reachability of most fasteners in the Mitsubishi Outlander battery pack, with few exceptions save for obstruction by other components or less reachable areas of the battery, such as those residing around the bottom casing. Fully autonomous and semi-autonomous tasks comprised 57% and 24% of total completion time, respectively, showing there is potential for significant productivity improvement given a range of repetitive tasks with readily identifiable components and where existing availability of tools facilitates fully autonomous task completion. The remaining 19% of tasks were classified as fully manual. Fig. 5 graphically shows various design features that were associated with lower autonomy levels. Interestingly, features associated with reduced disassembly cost, such as the use of adhesives and clips as an alternative to fasteners such as screws and bolts [6] were found to be associated with the increased complexity of automation, and vice-versa, suggesting a trade-off between these two factors when automation feasibility is incorporated into the disassembly process.

In Table 1, an estimated time for disassembly post-automation is recorded for tasks with the highest level of autonomy (A). We adjust the duration of each task of a given type k based on the efficiency factor, defined as

$$\eta_k = \frac{t_k^*}{t_k}, \quad (1)$$

where t_k , t_k^* are the required time per task instance for task type carried out manually and with a robot, respectively. For unbolting tasks classified as “A”, the estimated completion time post-automation is 11s, compared to 12s pre-automation, based on previous works [22], [23], [24], which addressed similar tasks. Based on these numbers, all tasks requiring unbolting were adjusted by a factor of 11/12 to estimate their post-automation time, reflecting the time it would take for a robot to complete the task. Conversely, tasks suitable for execution with a pick-and-place vacuum suction gripper were adjusted by a factor of 16.5/6.5. We apply this independently over each task instance i to provide a final estimate for task completion time post-automation, from which an overall efficiency factor η was derived as

$$\eta = \frac{\sum_i \eta_k t_{k,i}}{\sum_i t_{k,i}}. \quad (2)$$

The calculated final completion time for M2C disassembly post-automation, incorporating these adjustments, was 2708.8s, compared to 2741s pre-automation. This resulted in a efficiency factor of 1.01, rounded to 1, indicating that post-automation disassembling could be performed on the same number of battery packs within approximately the same time frame but requiring only one worker and one

TABLE 1. Table of main components and disassembly tasks for *pack-to-module* disassembly of the Mitsubishi Outlander PHEV battery pack, with tool requirements, completion times per task instance (to nearest half-second), and count of task instances per disassembly step. Aut. denotes the automation potential of the task, classified as autonomous (A), semi-autonomous (SA), or fully manual (M) processes. For “A” tasks, both the manual completion time (t_k [s]) and estimated robot completion time (t_k^* [s]) are reported.

Code	Component	Task	Tools	t_k [s]	t_k^* [s]	#	Aut.
1.1	Top case	Unbolt top case	Socket wrench	12	11	35	A
1.2		Remove top case	Hand	6.5	16.5	1	A
2.1	HV circuit (module side)	Remove terminal covers	Screwdriver, hand	4	—	10	M
2.2		Unbolt bus bar	Socket wrench	10	9	10	A
2.3		Remove bus bar	Hand	2	5	4	A
2.4 ¹		Re-fit terminal covers	Hand	6.5	—	8	M
3.1	Management harness (front)	Unclip management harness	Pliers	8	—	7	M
3.2		Unplug management harness	Screwdriver	4	—	2	M
4.1	HV circuit (junction box & service plug)	Unbolt HV harnesses	Socket wrench	15	14	2	A
4.2 ¹		Insulate HV harnesses	Hand	6.5	—	2	M
4.3		Unclip junction box covers	Pliers	5.5	—	7	M
4.4		Unbolt bus bars (junc. box side)	Socket wrench	13	—	4	SA ²
4.5		Remove bus bars	Hand	2	—	2	SA ²
4.6		Unbolt service plug	Socket wrench	6.5	6	2	A
4.7		Remove service plug	Hand	3	—	1	SA ³
4.8		Re-fit terminal covers	Hand	6.5	—	2	M
5.1	HV circuit (central)	Unclip HV harnesses	Screwdriver, pliers	3	—	6	M
5.2		Unclip central bus bar cover	Pliers	3.5	—	4	SA ⁴
5.3		Remove terminal covers	Screwdriver, hand	3.5	—	10	M
5.4		Unbolt bus bars	Socket wrench	10	9	10	A
5.5		Remove bus bars	Hand	3	7.5	4	A
5.6		Unbolt HV harnesses	Socket wrench	14.5	13.5	2	A
5.7		Remove HV harnesses	Hand	5	—	2	SA ³
6.1	Ventilation system	Unclip vent. system cover	Pliers	10	—	2	M
6.2		Unbolt ventilation system	Socket wrench	17	15.5	3	A
6.3		Unplug ventilation harness	Pliers	4	—	1	M
6.4		Remove ventilation system	Hand	5	—	1	SA ³
7.1	Management circuit	Unbolt heat exchanger cover	Socket wrench	10	9	4	A
7.2		Remove heat exchanger cover	Hand	2.5	—	1	SA ³
7.3		Unbolt rear bus bar	Socket wrench	13.5	12.5	2	A
7.4		Remove rear bus bar	Hand	3	7.5	2	A
7.5		Unplug management harness	Screwdriver	2.5	—	15	M
7.6		Unclip management harness	Pliers	5	—	25	M
7.7		Remove management harness	Hand	15	—	1	M
8.1	Rear junction box	Unclip rear junction box cover	Pliers	4	—	5	M
8.2		Unbolt bus bars	Socket wrench	13	12	6	A
8.3		Remove bus bars	Hand	4.5	—	3	SA ²
8.4		Unbolt resistor	Socket wrench	11.5	10.5	3	A
8.5		Remove resistor	Screwdriver, hand	5.5	—	1	A
8.6		Unbolt junction box	Socket wrench	16.5	15.5	2	A
8.7		Unplug junction box	Hand	2.5	—	1	M
8.8		Remove junction box	Hand	5.5	14	1	A
8.9		Unbolt junction box housing	Socket wrench	12	11	5	A
8.10		Remove junction box housing	Hand	5	—	1	SA ³
8.11		Unclip electronic module	Pliers, hand	12.5	—	2	M
9.1	Heat exchanger	Unbolt heat exchanger	Socket wrench	11	10	3	A
9.2		Unbolt heat exchanger plug	Socket wrench	17	—	4	SA ⁵
9.3		Remove heat exchanger	Hand	5	—	1	SA ³
10.1	Front junction box	Unbolt junction box housing	Socket wrench	13.5	12.5	7	A
10.2		Unbolt LV harnesses	Socket wrench	15	14	3	A
10.3		Remove junction box housing	Hand	5	—	1	SA ³
11.1	Module stacks	Unbolt module cover plates	Socket wrench	20	18.5	8	A
11.2		Remove module cover plates	Hand	3	7.5	3	A
11.3		Unbolt module stack	Socket wrench	23.5	—	20	SA ²
11.4		Remove module stack	Hand	2.5	6.5	5	A
11.5		Unbolt exterior plugs	Socket wrench	13.5	12.5	11	A

¹ Performed for safety, but not a physical prerequisite for other tasks

² Possibility for occlusion/obstruction by other components

³ Suction gripping or grasping may fail due to complex geometry or weight distribution about grasp points

⁴ With robot-mounted specialized tool

⁵ Low visibility/reachability

robot, maintaining similar processing levels. However, labor productivity, defined as output per hour worked, would nearly

double, due to reduction of labor-intensive processes as Fig. 4.

TABLE 2. Table of main components and disassembly tasks for *module-to-cell* disassembly of the Mitsubishi Outlander PHEV battery pack, with tool requirements, completion times per task instance (to nearest half-second), and count of task instances per disassembly step. Aut. denotes the automation potential of the task, classified as autonomous (A), semi-autonomous (SA), or fully manual (M) processes. For “A” tasks, both the manual completion time (t_k) and estimated robot completion time (t_k^*) are reported.

Code	Component	Task	Tools	t_k [s]	t_k^* [s]	#	Aut.
1.1	Module stack ¹	Remove top case	Screwdriver, hand	18	—	1	SA
1.2		Remove modules ²	Screwdriver, hand	6	—	2	SA
2.1	Module ³	Unclip top cover	Screwdriver	6.5	—	1	M
2.2		Unplug management circuit	Screwdriver	6	—	1	M
2.3		Remove terminal covers ⁴	Screwdriver, hand	2.5	—	2	M
2.4		Unbolt cell terminals	Socket wrench	16	14.5	14	A
2.5		Remove cell connector cable	Hand	12	—	1	M
2.6		Remove bus bars	Pliers	4	—	7	SA ⁵
2.7		Unclip cell housing cage ²	Screwdriver	80.5	—	1	M
2.8		Unclip management circuit	Pliers	2.5	—	1	M
2.9		Unclip cell housing cover	Pliers	5	—	1	M
2.10		Remove cells	Pliers, hand	3	—	1	SA ⁵

¹ Five stacks per pack; each task is defined *per stack*.

² Destructive to the mounting/housing component.

³ Two modules per stack; each task is defined *per module*.

⁴ Reverses task 6.8 of pack-to-module procedure in Table I.

⁵ Automation is possible by removing via gravity.

B. MODULE-TO-CELL DISASSEMBLY

In the AccurecTM process, pack-to-module disassembly is carried out for EV batteries, followed by thermal treatment, shredding and sorting [25]. Disassembly beyond module level, i.e. *pack-to-cell*, may optionally be carried out prior to further processing. For the Outlander case study, individual module-to-cell (M2C) disassembly was similarly time-consuming to the entire pack-to-module process (775s per module stack, containing two modules). From Fig. 4c-4d, the majority of time spent on module-to-cell disassembly was on unbolting, a potentially automatable task. Furthermore, around 2/3 of the disassembly completion time was spent on semi-autonomous or fully-autonomous tasks. For robotizing the fully autonomous tasks, the per-stack M2C disassembly time was reduced to 738.6s, resulting in a efficiency factor of $\eta = 1.05$. However, the M2C process has comparatively greater time spent on fully manual processes than for P2M disassembly, indicating the higher complexity of the tasks required for M2C. By far the most time-consuming manual task was the removal of the cell housing cage (Table 2, task 2.7), which required unclipping of numerous unmarked fasteners and is thus challenging to identify autonomously due to the lack of distinctive visual features indicating their presence. Even with the presence of non-destructively removable and distinctive fasteners - a factor uncommon in module design - these trends corroborate existing difficulties in M2C disassembly [6], [26].

Whereas time-consuming, automation of these tasks represents a weighty challenge with non-destructive methods; furthermore, given the semi-destructive nature of the existing manual process, and the comparatively low value of the housing components, it is of interest to develop alternative, semi-destructive approaches such as cutting [13]. These are distinct from fully destructive practices such as shredding by allowing mechanical separation of the casing without damage to the active materials of the battery. For these processes,

process planning is a difficulty for disassembly applications where the robot must cut materials without prior knowledge of the specific materials in question [27]. However, this area of research has been sparsely explored, and thus, developing methods suitable for general application to M2C disassembly remains an active research problem. These challenges are compounded by unique safety considerations for battery disassembly in the risk of damaging the sensitive battery components within and risk of thermal runaway.

IV. ECONOMIC EVALUATION

Whereas in Section III, task feasibility considerations imply necessity of human-robot collaboration, different types of collaborative robots (“cobots”) could be used in industry for the robotizing of the workcell for the disassembly process. These robots’ primary limitation lies in the carrying capacity they can support and supported sensory modalities, such as measuring external force/torque, and control modalities (position vs effort-based control), which influence the tasks that can be carried out. In the process, we assume that the task can be completed if the robot carrying capacity exceeds the weight of the heaviest battery component. Common collaborative robots and their specifications are underlined in Table 3. Our modeling scenario is based on the costs for the German company Accurec Recycling GmbH, as a basis for the robotization of the disassembly process. For our analysis, we adopt values for the wages of a disassembly technician as approx. 25€/hr, electricity costs as 0.418€/KWh [28] and the recovery per kg of battery as 1.8€/kg.¹ However, we note the subsequent analysis is generally applicable across geographic regions given local values of labor and utility costs are known.

¹Based on consultation with an Accurec engineer (personal communication, October 2023).

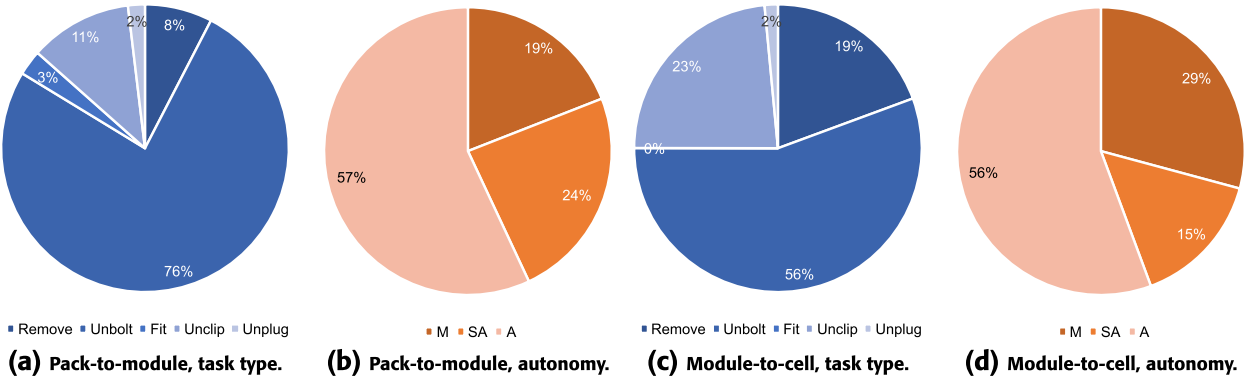


FIGURE 4. Breakdown of pack-to-module and module-to-cell disassembly completion times by type of task and estimated level of autonomy.

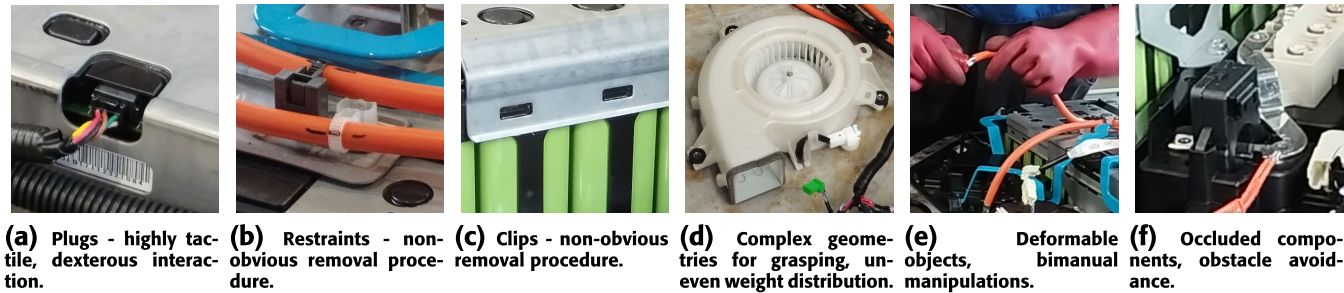


FIGURE 5. Use of design features in the Mitsubishi Outlander PHEV battery pack associated with lower levels of autonomy (Table 1, 2).

TABLE 3. Comparison of commercially available collaborative robots with their rated payload (carrying capacity), power consumption and maintenance costs during typical operation.

Robot manufacturer	Price range [€]	Carrying capacity	Annual amortized cost [€/yr]	Maintenance cost [€/yr]	Power consumption [W]	DBT/1y RoI ¹
FANUC	10000–50000	4 to 30kg	1666–8333	516–2580	100–500	1–2
Universal Robots	10000–50000	UR10e 12.5kg	1666–8333	516–2580	350	1–2
		UR16e 16kg			350	
		UR20 20kg			500	
KUKA	100000+	LBR iiwa 7kg	16666+	5160+	350	4+
		LBR iiwa 14kg			500	
		Other med-sized			425	

¹ Required daily battery throughput for return on investment (RoI) within 1 year

A. FORMULATIONS

We consider the case study of adapting a setup consisting of six disassembly cells with two disassembly technicians per cell as per procedures set out in II-B and current manual disassembly practices, which we label the “manual” scenario. In a similar vein to other proposed disassembly concepts [11], while acknowledging the prevalence of manual tasks in the disassembly process, we propose a disassembly cell architecture consisting of six cells with one disassembly operator and one cobot. We label this scenario the “robotized” scenario. In this scenario, we assume that the robot and human technician work concurrently on tasks appropriate for their level of autonomy, which implies the allowance for nonlinearity in the disassembly process from Table 1, 2. We posit this assumption is reasonable due to the

high percentage of fully autonomous tasks for both P2M and M2C disassembly, as well as their relatively uniform dispersal throughout the P2M process.

Regarding productivity gains in the modeling scenario, as of January 2024, the average weekly working hours in Germany are 36.7, totaling around 1,800 hours annually when accounting for public holidays and leave [29]. In contrast, a robot in a disassembly plant could theoretically operate 24/7, potentially reaching 8,700 working hours annually under ideal conditions. However, this assumes minimal downtime for maintenance, safety checks, and other standard operating procedures. In practice, robot operation hours may be limited by factors such as the need for human supervision, partially automated processes requiring human intervention, and plant closures during non-working hours.

TABLE 4. Quantities used in economic evaluation of robotizing disassembly process.

	Description	Unit	Value
P_c	Electricity cost, Germany [28]	€/kWh	0.418
E_A	Energy cost, robot	€/year	319.77
C	Purchase price, robot	€	28304.4
t_{RL}	Useful life, robot	years	6
V_C	Recovery value of cell	€/kg	1.8
β	Cell fraction of battery weight	—	0.4
W	Battery pack mass	kg	150
N_B	Throughput per worker, manual process	#/day	9
η	Robot efficiency factor (Sec. IV)	—	1.01
N_e	Number of disassembly technicians	—	6
c_e	Cost of wages per employee	€/day	€200
t_y	Working days, annually	days/year	245
t_s	Salaried days, annually	days/year	260

Therefore, to simplify the subsequent analysis, changes in the operational schedule due to automation are considered as beyond the scope of this work, and thus we assume that the total number of operating hours remains unchanged.

Considering human and robot operating costs, assuming the power consumption of the average of 425W for the KUKA LBR iiwa and Universal Robots, and change in throughput as (2), the annual profitability for the “robotized” scenario can be calculated as follows:

$$I_A = N_e [(1 + \eta) V_B N_B t_y - c_e t_s - c_r t_y], \quad (3)$$

where c_e , c_r are the daily cost per employee and robot, respectively, and

$$V_B = V_C \cdot \beta W \quad (4)$$

is the net recovery value per battery, with the efficiency factor η defined as (2) and quantities derived in Table 4. The change in annual profitability from the manual scenario is therefore

$$\Delta I_A = N_e [(\eta - 1) V_B N_B t_y + c_e t_s - c_r t_y]. \quad (5)$$

and thus, the payback period can be expressed in terms of the return on investment (RoI) period in years

$$t_{RoI} = \frac{C \cdot N_e}{\Delta I_A}. \quad (6)$$

B. COST ESTIMATION

The initial cost of a collaborative robot can vary widely depending on its capabilities, size, and the specific tasks it is designed to perform. Prices can range from as low as \$10,000 for simple, small robots to over \$100,000+ for large, complex systems. This cost typically includes the robot, necessary peripherals, and sometimes initial setup, programming and training courses. Maintenance costs for industrial robots are a critical consideration for any business planning to utilize this technology. These costs can be divided into several categories, as summarized in Table 5. A common industry guideline suggests annual maintenance costs for industrial robots range from 3% to 8% of the initial purchase price. This is based on data from various robot manufacturers and industry reports, though exact figures can vary [6], [30].

In (6), the required RoI period to cover the initial robot cost is considered. To amortize the robot cost (C) over its useful life, we divide C by the robot expected useful life (in years) to determine the annual amortization cost. This annual cost replaces the total purchase price in the payback equation. Additionally, we include recurring operating costs such as estimated annual maintenance and consumables (e.g., specific end-effectors). First, the annual maintenance (AM) cost of the robot is estimated as:

$$AM = \frac{\min + 4\mu + \max}{6}, \quad (7)$$

where $\min$, μ , and $\max$ are the minimum, mean, and maximum estimated annual maintenance costs, respectively. Using PERT estimation, we take a weighted average that gives greater importance to the most likely cost μ , which makes it more accurate than a simple average of the three points. Next, the annual amortized (AA) robot cost is computed as:

$$AA = \frac{C}{t_{RL}}, \quad (8)$$

where C is the total robot cost and t_{RL} is the robot useful life in years. The total annual amortized robot cost is thus

$$c_{r,A} = AA + AM + E_A. \quad (9)$$

where E_A is the annual electricity usage cost. The daily robot cost c_r follows similarly as

$$c_r = \frac{c_{r,A}}{t_y}. \quad (10)$$

The number of battery packs (N_A) needed annually to cover the robot cost is then given by:

$$N_A = \frac{c_{r,A}}{V_B}, \quad (11)$$

To calculate the capital cost of robot investment, we employ the nominal annuity depreciation method (i.e., linear capital cost during the asset’s economic lifetime). This approach considers not only the initial cost of the robot but also factors such as financing, depreciation, and expected return on investment. The capital cost is calculated as follows:

$$CC = C_0 * \frac{\tau(1 + \tau)^{t_{RL}}}{(1 + \tau)^{t_{RL}} - 1} \quad (12)$$

where CC represents the capital cost in euros (€), C_0 is the principal amount invested, also known as the initial cost, in euros (€), τ represents the discount rate as a percentage (%). This formula allows businesses to estimate the true cost of robot ownership over time, accounting for both the initial investment and the expected returns, adjusted for risk and inflation [26].

TABLE 5. Maintenance costs for industrial robots.

Category	Description	Cost Considerations
Regular Maintenance	Routine inspections and replacement of wear-and-tear parts	Lower cost; Prevents breakdowns and extends robot lifespan
Corrective Maintenance	Repairs and replacements after failures	Higher cost; Includes potential production downtime
Software Updates	Keeping software up-to-date for optimal performance and security	Costs include subscriptions or one-time fees for updates
Training and Support	Training staff to operate and maintain the robot, ongoing support	Initial training costs, ongoing education, support contracts

C. COST-BENEFIT ANALYSIS (CBA)

We use the net present value (NPV) method to assess the profitability of investment in robotic disassembly, defined as

$$NPV = -CC + \sum_{n=1}^{t_{RL}} \frac{R_n}{(1 + \tau)^n}, \quad (13)$$

where n represents the time period in years, R_n denotes the cash flow at time n in euros, and τ is the discount rate expressed as a percentage. A positive NPV indicates a favorable investment. For our analysis, we consider a case study where $N_c = 6$ disassembly cells are adapted to the automated disassembly concept described in Section IV-A. This involves replacing one worker in each cell with a collaborative robot. The initial investment cost (C_0) is calculated as $N_c C = €169,826.40$ based on the cost of six collaborative robots, as determined by our market research for suitable models in this application. The cash flows R_n for each year are estimated using (5), with annual costs from (9), excluding amortized cost AA. Based on industry data and our case study analysis, we assume an annual throughput of €2,860,381.47 and a first-year operational cost of €351,003.04. Additionally, we project a 10% increase in robot efficiency for the first three years due to improved disassembly line balancing. This improvement is based on optimization techniques applied to robotic disassembly, as demonstrated in previous studies [31].

We calculate the capital cost using (12), with a discount rate (τ) of 12%. This rate is chosen conservatively reflecting the expected return on investment adjusted for risk, based on industry standards for similar automation projects [32], [33]. Applying the NPV calculation (13) for the robotized scenario yields an NPV of €2,218,187.15. This positive NPV suggests that the investment in robotic disassembly is highly favorable and expected to generate significantly more value than the initial investment cost over the project's lifetime. Notably, however, this is predicated on the assumption of a uniform service life of 6 years for all hardware, consistent market conditions for recovered materials, and no major technological disruptions during the investment period. While these assumptions provide a solid basis for decision-making, actual results may vary based on specific implementation details and market fluctuations. Hence, regular reassessment of the project's economic performance

TABLE 6. Financial summary of robotic integration of battery disassembly under the scenario of 6 employees with 1 Universal Robot per employee and nominal productivity change scenario ($\eta = 1.01$).

Description	Amount [€]
Annual amortized cost and robot operating cost (6 robots)	39,003.04
Annual salary for six technicians	312,000.00
Total annual costs	351,003.04
Annual revenue with material recovery	2,874,683.38
Net annual revenue (with robots)	2,523,680.33
Net present value (with robots)	2,288,359.03
Net annual revenue (traditional)	2,236,381.47
Decrease in disassembly cost	43.75%
Increase in net annual revenue	12.85%

is recommended to ensure continued alignment with business objectives.

The case study of a human-robot collaborative workstation in a disassembly outlet demonstrates significant financial and operational benefits in a two-operator work cell. By upgrading the work cells, where each now employs one technician and one Universal Robot rather than two technicians, the operational dynamics have been substantially optimized without reducing the daily disassembly throughput. A daily production rate of 9 batteries is estimated based on disassembly times established in earlier sections, considering working hours, the number of cells, and current manual disassembly throughput. Despite the unchanged daily production rate, the financial implications are profound, as summarized in Table 4-C. The introduction of Universal Robots maintains efficiency while significantly reducing labour costs. Each robot incurs an annual operating cost of €6500.51 (from (8)), leading to a total annual robot operating cost of €39,003.04 for six robots. When combined with the reduced labor costs, where only six technicians are needed (reduced from twelve), the total labor costs amount to €312,000 annually, substantially lower than the €624,000 required previously.

This strategic deployment of automation facilitates an annual revenue from the material recovery of €2,860,381.47, derived from processing 9 battery packs per day. Consequently, the net annual revenue with the robotic system is €2,509,378.43, a significant improvement over the €2,236,381.47 from the traditional method, marking a 12.85% increase. The total cost savings amount to a 43.75% reduction in disassembly costs, illustrating the economic

efficiency of integrating Universal Robots into traditionally labor-intensive processes.

V. ROADMAP TOWARDS AUTOMATED DISASSEMBLY

While automation of certain tasks can boost revenue, the process would still rely heavily on human-robot collaboration. Previous research, as the paper [6] by Lander et al. outlines a roadmap towards full autonomy, however, is predicated on the replacement of manual tasks with automatable tasks. Some processes in the battery disassembly have already been fully automated in research carried out in the robotics domain. Table 1 and Table 2 provide a detailed assessment of the tasks that are carried out in the disassembly of the Mitsubishi Outlander PHEV battery pack. To further contribute in the analysis of the disassembly potential, some of the fully autonomous (A), semi-autonomous (SA) and manual (M) tasks carried out in related work [5], [9], [13], [22], [23], [24], [27], [34], [35], [36], [37] are used as reference, also helping with the analysis carried out in Section II-C. With this previous work, we can confidently argue the automation potential of the tasks.

A. EFFECT OF AUTONOMY OF EV BATTERY DISASSEMBLY

To consider the effect of task feasibility on disassembly cost and revenues, we aggregate task feasibility into a metric of *autonomy* over the disassembly process, defined as

$$\alpha_{\max} = \frac{t_A + \gamma t_{SA}}{t} \quad (14)$$

where t is the total disassembly time, t_A is the sum of time spent on “A” tasks over each task, and similarly for “SA” tasks (t_{SA}). γ is a factor that aggregates the overall level of autonomy of semi-autonomous tasks (here, we assume this is constant over all components) concerning a fully autonomous task - equivalently, it encodes the ratio of additional time required to perform an SA task due to human interventions, task failures or repeated task attempts. For a given number of human, robot workers N_e , N_w , the *physical* disassembly autonomy arises from (14) as the fraction of the baseline disassembly time carried out by robot workers,

$$\alpha = \frac{\eta N_r}{N_e + \eta N_r}. \quad (15)$$

To examine the effect of autonomy on disassembly cost and output, we consider the ratio of automatable to manual tasks informs the required ratio of the robot to human workers

$$v = \frac{N_r}{N_e + N_r} \quad (16)$$

where v , α are related as:

$$v(\alpha) = \frac{\alpha}{\alpha + \eta(1 - \alpha)} \quad (17)$$

and in the optimal case, robot resources are allocated such that all automatable tasks can be carried out by robot workers with minimal downtime, and thus

$$\alpha = \alpha_{\max}. \quad (18)$$

Hence, $\alpha_{\max}$ governs the allocation of tasks between human and robot workers. Note that given a single disassembly workstation, not all values of v (and hence α) are physically realizable, and the true value of α will be aliased to that of common configurations, e.g., 0.5 for 1 robot, 1 human, 2/3 for 2 robots, 1 human, given $\eta = 1$. Nonetheless, we posit that human and robot workers can be allocated across multi-workstation setups such that the equality (18) holds. However, for completeness, we consider this case in Appendix.

To relate the task feasibility to overall disassembly profitability, we formulate the dependence of disassembly revenue as a function of the number of human and robot workers as follows:

$$B = (N_e + \eta N_r) V_B N_B \quad (19)$$

next, we parameterize (19) in terms of v and the total number of workers:

$$N_w = N_e + N_r, \quad (20)$$

and thus using (16), (20), total disassembly revenue is

$$B = N_w (1 + v(\eta - 1)) V_B N_B, \quad (21)$$

while disassembly cost per worker follows similarly as

$$c_w = (1 - v)c_e + v c_r. \quad (22)$$

Thus, the overall profitability of the disassembly process can be parameterized as a function of η , v :

$$I = N_w [(1 + v(\eta - 1)) V_B N_B - c_w]. \quad (23)$$

Considering (23), the break-even points versus manual disassembly and overall financial break-even can be computed. In the former case, automation at a given level of α is expected to outperform manual disassembly ($\alpha = 0$) when the following criterion is met

$$\eta = 1 + \frac{c_r - c_e}{V N_B}, \quad (24)$$

while overall financial break-even in the case that manual disassembly is unprofitable is considered as

$$\eta \geq 1 + \frac{c_w - V N_B}{v V N_B}. \quad (25)$$

Equivalently,

$$\eta \geq \frac{\alpha c_r}{V N_B - (1 - \alpha) c_e}. \quad (26)$$

Fig. 6 shows the dependence of (23) with respect to α and η under the base modeling scenario. Increasing autonomy and efficiency are indicative of increasing profitability, with the direction of greatest change indicating revenues are primarily dependent on increasing efficiency factor in the near-term, while increasing level of automation in the near-term has marginal effect. However, in the long-term, diminishing returns on increasing efficiency factor favor a long-term strategy of increasing autonomy.

Fig. 7 shows the required efficiency factor to satisfy the break-even criteria (24), (26) with varying recovery values per cell. These are indicated for the base modeling scenario and a reduced throughput scenario of $N_B = 2$ battery packs per worker per day. The base scenario corroborates the overall profitability dependence in Fig. 6, with break-even being dependent on the manual process (24) and no dependence of overall break-even on autonomy, while as recovery values per cell decrease, correspondingly lower efficiency factors are tolerated. On the other hand, the dependence of autonomy becomes more notable with decreasing value of battery material, and particularly for the lower throughput scenario, where overall financial break-even (26) becomes significant. The plots suggest two primary application areas for robotics in EV battery disassembly:

- Recycling of high-value materials in a human-robot collaboration setting. Robots assist with a few tasks with selectively high efficiency factors. Robots primarily increase operating revenues.
- Recycling of low-value materials with low base throughput. Robots assist throughout the disassembly process, and may be slower than humans at carrying out tasks. Robots primarily reduce operating costs.

Wherein the latter application implies the role of robotics for disassembly of non-serviceable or damaged batteries, where robots are ideally suited to carry out safety-critical tasks. In particular, with reducing recovery value per cell V_C , the required level of automation correspondingly increases. Therefore, commercial viability of disassembly for recycling of complex or damaged batteries under this scenario is predicated on increasing level of automation and adaptation to uncertain product conditions, in contrast to improving task efficiency for repetitive, simple tasks.

B. REPLACEMENT OF MANUAL TASKS

Tasks with good reachability and identifiable characteristics are suitable for common end-effectors and tools. Based on consideration of previous research developments in the area of robotic disassembly, if a disassembly task meets similar requirements to another task, an extrapolation can be made, and with some adjustments, the tasks should be able to be robotized. From this, we propose candidate replacement tasks (Table 7, 8 for P2M and M2C, respectively) for existing manual and semi-autonomous processes and the necessary robot tools to complete the disassembly task successfully, motivated by the illustrative examples presented in Fig. 8. These examples demonstrate several capabilities relevant to the process of battery disassembly.

Two-finger grippers, used in teleoperated or SA modes, are effective for various tasks, including bus bar and cable management, as demonstrated in Fig. 8a, 8f. This gripper type could also facilitate the removal of components, such as service plugs, heat exchangers, and management harnesses, among others listed in Table 7, 8. Two-finger grippers handle larger, heavier objects in teleoperated or SA modes,

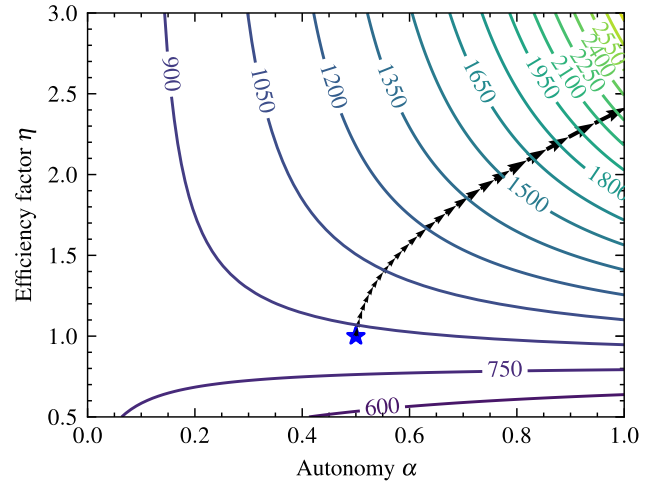


FIGURE 6. Contour plot of disassembly profitability per worker as a function of the level of disassembly automation α and robot efficiency factor η under the considered disassembly case study. The estimated values of α and η for the Mitsubishi Outlander PHEV with current disassembly tasks are shown (*). The trajectory of the greatest change in profitability (arrows) indicates the primary focus area for increasing automation benefits over time.

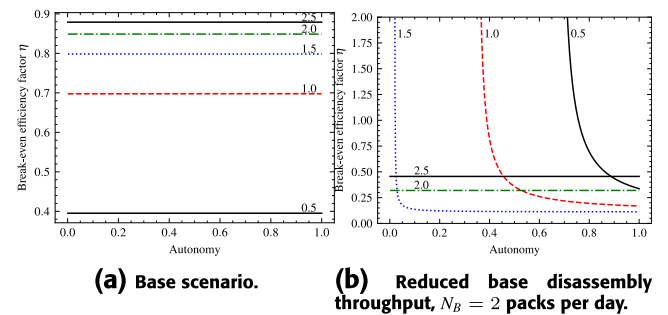
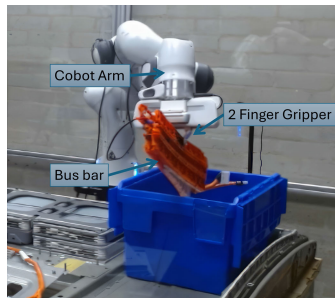


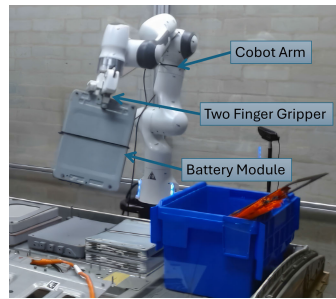
FIGURE 7. Plot of required robot efficiency factors and level of disassembly automation for twofold break-even under the considered disassembly case study, with changing recovery values per cell (€/kg).

enhancing maneuverability and force application. Fig. 8b, 8k shows their use in moving battery modules and covers. This tool applies to other disassembly tasks, such as ventilation systems and junction boxes, benefiting from the gripper's capabilities. Vacuum suction grippers handle large or heavy objects, enhancing robotic arms tasked with heavier loads, as shown in Fig. 8g, 8h. Smaller grippers assist in confined spaces, facilitating the removal and fitting of components like terminal covers and battery cells.

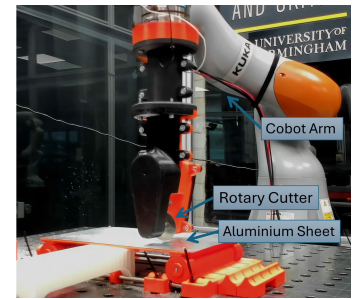
Socket wrenches, often automated with vision algorithms, require semi-autonomous control for complex geometries. Fig. 8i, 8j illustrate their use in bolt removal and transportation. For non-removable fasteners, rotary cutters and other cutting tools mounted on collaborative robots facilitate disassembly tasks like removing harnesses and cutting casings, as shown in Fig. 8c, 8d. These tools can also be adapted for semi-autonomous operation across various tasks, such as cutting of cables, object casings and housings, which often can require better depth control.



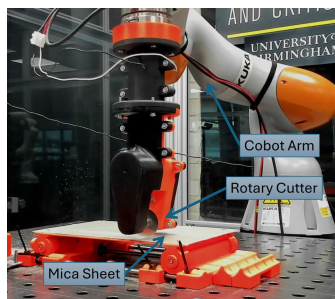
(a) Teleoperated (SA or M) busbar removal and transportation from the battery pack, using a 2-finger prehensile gripper.



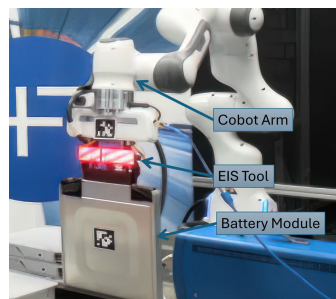
(b) Grasping and removal of battery module casing with 2-finger prehensile gripper (SA or M).



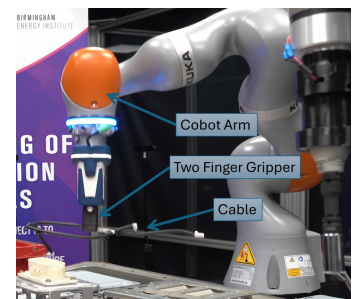
(c) Teleoperated or autonomous (A, SA or M) aluminium sheet cutting. A rotary cutter was used to cut the sheets to demonstrate cutting metallic housing components.



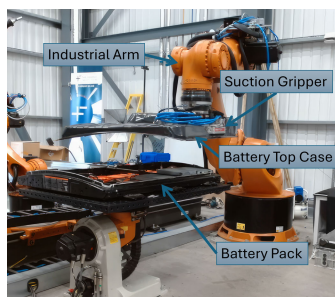
(d) Teleoperated or autonomous (A, SA or M) mica sheet cutting. A Rotary Cutter was used to cut the sheets to demonstrate cutting battery module pouches.



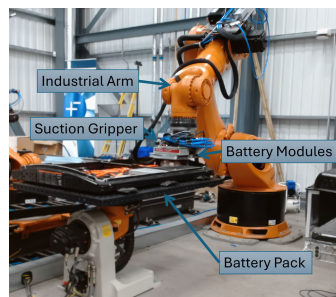
(e) Vision guided electrochemical impedance spectroscopy (EIS, A or SA). The robot connects to battery terminals to perform state of health analysis.



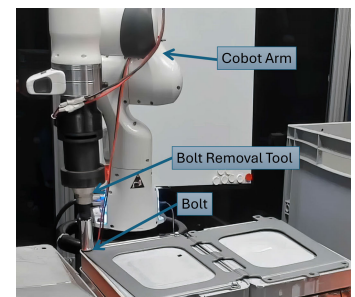
(f) Grasping and removal of battery harness connector with 2-finger prehensile gripper (SA or M).



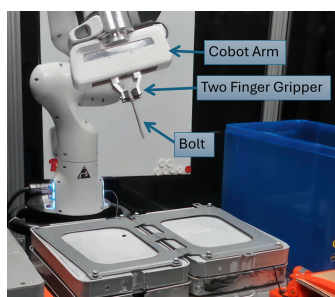
(g) Autonomous top case removal (A or SA) with industrial robot and vacuum suction gripper.



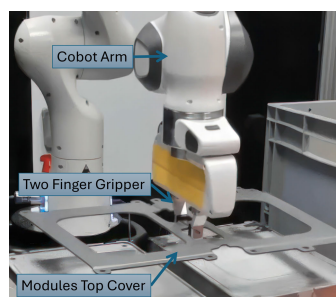
(h) Sorting of Battery stacks in industrial setup (A or SA). The industrial robot performs battery module sorting utilizing a vacuum suction gripper.



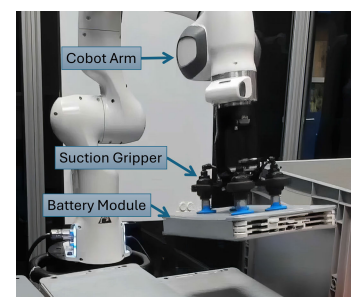
(i) Teleoperated or autonomous (A, SA or M) bolt removal. The robot uses a motorized socket wrench to remove the bolts around the battery module.



(j) Bolt removal with 2-finger prehensile gripper (A, SA, or M).



(k) Cover removal with 2-finger prehensile gripper (A, SA, or M).



(l) Module sorting with vacuum suction gripper (A, SA or M).

FIGURE 8. Tasks performed by robots related or applicable to battery disassembly. The tasks are performed across a variety of workspaces and are either autonomous (A), semi-autonomous (SA) or manual via teleoperation (M), but serve as a basis for the automation of the whole battery pack disassembly.

TABLE 7. Main components and candidate replacement disassembly tasks for *pack-to-module* disassembly of the Mitsubishi Outlander PHEV battery pack, with proposed solutions and proposed robot tool for completing the task. Aut. denotes the automation potential of the original and revised task, classified as autonomous (A), semi-autonomous (SA) or manual (M).

Code	Component	Task	Robot Tools	Proposed Solution	Aut.
2.1	<i>HV circuit (module side)</i>	Remove terminal covers	Suction gripper	Similar to Fig. 8l	M→SA
2.4		Re-fit terminal covers	Suction gripper	Similar to Fig. 8l	M→SA
3.1	<i>Management harness (front)</i>	Separate mgmt. harness restraints	Rotary cutter tool	Similar to Fig. 8c & 8d	M→SA
3.2		Separate mgmt. harness plug	Shears	Similar to Fig. 8c & 8d	M→SA
4.2	<i>HV circuit (junction box & service plug)</i>	Insulate HV harnesses	N/A		M
4.3		Separate junction box covers	Rotary cutting tool	Similar to Fig. 8c & 8d	M→SA
4.4		Unbolt bus bars (junc. box side)	Socket wrench	Similar to Fig. 8i & 8j	SA
4.5		Remove bus bars	Suction gripper	Similar to Fig. 8g & 8h	SA
4.7		Remove service plug	Prehensile gripper	Similar to Fig. 8a & 8f	SA
4.8		Re-fit terminal covers	Suction gripper	Similar to Fig. 8l	M→SA
5.1	<i>HV circuit (central)</i>	Separate HV harness restraints	Rotary cutting tool	Similar to Fig. 8c & 8d	M→SA
5.2		Separate central bus bar cover	Rotary cutting tool	Similar to Fig. 8c & 8d	SA
5.3		Remove terminal covers	Suction gripper	Similar to Fig. 8l	M→SA
5.7		Remove HV harnesses	Prehensile gripper	Similar to Fig. 8a & 8f	SA
6.1	<i>Ventilation system</i>	Separate vent. system cover	Rotary cutting tool	Similar to Fig. 8c & 8d	M→SA
6.3		Unplug ventilation harness	Shears	Similar to Fig. 8c & 8d	M→SA
6.4		Remove ventilation system	Prehensile gripper	Similar to Fig. 8b & 8k	SA
7.2	<i>Management circuit</i>	Remove heat exchanger cover	Suction gripper	Similar to Fig. 8g & 8h	SA→A
7.5		Separate mgmt. harness plug	Shears	Similar to Fig. 8c & 8d	M→SA
7.6		Separate mgmt. harness restraints	Rotary cutting tool	Similar to Fig. 8c & 8d	M→SA
7.7a		Separate management harness	Shears	Similar to Fig. 8c & 8d	SA
7.7b		Remove management harness	Prehensile gripper	Similar to Fig. 8a & 8f	M→SA
8.1	<i>Rear junction box</i>	Separate rear junction box cover	Rotary cutting tool	Similar to Fig. 8c & 8d	M→SA
8.3		Remove bus bars	Prehensile gripper	Similar to Fig. 8a & 8f	SA
8.7		Separate junction box plug	Shears	Similar to Fig. 8c & 8d	M→SA
8.10		Remove junction box housing	Prehensile gripper	Similar to Fig. 8b & 8k	SA
8.11		Separate electronic module	Rotary cutting tool	Similar to Fig. 8c & 8d	M→SA
9.2	<i>Heat exchanger</i>	Unbolt heat exchanger plug	Socket wrench	Similar to Fig. 8i & 8j	SA
9.3		Remove heat exchanger	Prehensile gripper	Similar to Fig. 8a & 8f	SA
10.3	<i>Fr. junc. box</i>	Remove junction box housing	Prehensile gripper	Similar to Fig. 8b & 8k	SA
11.3	<i>Mod. stacks</i>	Unbolt module stack	Socket wrench	Similar to Fig. 8i & 8j	SA

TABLE 8. Main components and candidate replacement disassembly tasks for autonomous *module-to-cell* disassembly of the Mitsubishi Outlander PHEV battery pack, with proposed solutions and proposed robot tool for completing the task. Aut. denotes the automation potential of the task, classified as autonomous (A), semi-autonomous (SA) or manual (M).

Code	Component	Task	Robot Tools	Proposed Solution	Aut.
1.1	<i>Module stack</i>	Remove top case	Suction gripper	Similar to Fig. 8g & 8h	M→SA
1.2		Remove modules	Suction gripper	Similar to Fig. 8g & 8h	M→SA
2.1	<i>Module</i>	Separate top cover	Rotary cutter tool	Similar to Fig. 8c & 8d	M→A
2.2		Separate management cable	Shears	Similar to Fig. 8c & 8d	M→SA
2.3		Remove term. covers	Suction gripper	Similar to Fig. 8l	M→SA
2.5		Remove cell connector cable	Prehensile gripper	Similar to Fig. 8a & 8f	M→SA
2.6		Remove bus bars (gravity)	Suction gripper	Similar to Fig. 8g & 8h	SA
2.7		Cut cell housing cage	Rotary cutter tool	Similar to Fig. 8c & 8d	M→SA
2.8		Unclip management circuit	N/A		M
2.9		Remove cell housing cage	Prehensile gripper	Similar to Fig. 8b & 8k	M→SA
2.10		Remove cells	Suction gripper	Similar to Fig. 8l	SA

In Table 7, a majority of manual tasks were automatable by changing the nature of the task, for example, replacing unclipping or unplugging tasks with semi-destructive mechanical separation, typically of the covers or casings for various components. Only one fully manual task could not be automated or made semi-autonomous, which was the insulation of the HV harnesses in step 4.2. This is due to the interaction with deformable objects, which requires complex, bimanual manipulations in combination with high amounts of force to insert the loose ends of the harness into the insulation medium. Predominantly, the semi-autonomous tasks could not be promoted to fully autonomous due to the need to define appropriate positions for cutting, separating, and grasping.

These components lack clear visual cues that would allow for autonomous planning of these tasks; therefore, some prior knowledge or user intervention would be required to accomplish the task. Prior knowledge may be insufficient for deformable objects due to the uncertain position of the objects between individual battery packs. Few tasks could be promoted to fully autonomous, indicating significant barriers remain in bridging the autonomy gap between semi-autonomous and fully autonomous task execution, which is crucial for the realization of a fully automated battery disassembly process.

Key outstanding issues persist across various manipulation and cutting tasks. In robotic cutting, for instance, difficulties

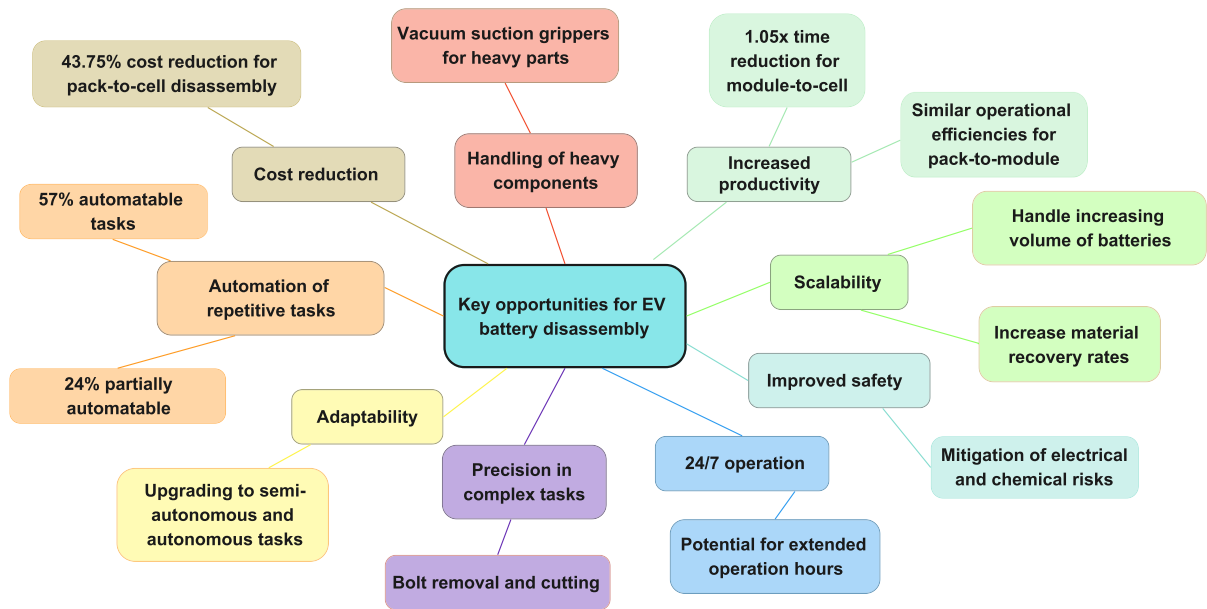


FIGURE 9. Illustration of main opportunities for integrating robotics into electric vehicle (EV) battery disassembly processes. These include cost reduction, automation of repetitive tasks, increased operational efficiency, improved safety, scalability for growing demand, 24/7 operation potential, handling of heavy components, precision in complex tasks, and adaptability for semi-autonomous processes.

arise in the reliable identification of fasteners or restraining elements, the planning of efficient cutting paths [38], and process planning on unknown or variable materials, which could impact tool wear and the integrity of surrounding components. In tasks involving prehensile grasping, manipulation, or the use of shear tools, segmentation and grasp planning remain challenging, particularly with respect to selecting graspable objects, determining stable grasp positions, and avoiding collisions. Similarly, for suction grasping, the identification of suitable flat surfaces is a key limitation, as such areas may be rare or obstructed by irregular component shapes or deformations. Each of these tasks would benefit from advancements in segmentation and component recognition to improve automation capabilities. Furthermore, this highlights the necessity of human-robot collaboration and “human-in-the-loop” decision-making in the near term to accomplish disassembly.

C. SUMMARY

In the current context of task feasibility and the automatability of industrial processes, it is evident that integrating collaborative robots (cobots) into work cells can lead to a marked reduction in costs. Specifically, there is a notable decrease in disassembly costs by 43.75%. This reduction is significant, particularly in light of the increasing volume of waste and the higher proportion of lithium-ion batteries from electric vehicles reaching their end of life (EoL), escalating the volume of waste to be processed. Fig. 9 highlights the main advantages of robotics in EV battery disassembly, including cost reduction, improved efficiency, safety, and scalability.

Current literature on techno-economic analysis, such as [26] and [39], often overlooks selective efficiency

factors. That is, they fail to consider which tasks are currently amenable to automation. Presently, automation primarily encompasses the removal of the battery case. The literature tends to apply a singular efficiency factor that includes the robotic disassembly of the entire pack-to-module sequence. This approach is challenging to calculate accurately, as specific tasks still require human operators to be involved. Improvement in disassembly line balancing [31], [40], and the manufacturing industry’s aspiration to maximize efficiency and productivity suggests that future work cells may exclusively comprise cobots. These cobots would operate in unison to perform sequential tasks in an automated manner. This shift implies that we would not be constrained by the operating hours currently aligned with technicians’ working hours, which are approximately 1800 hours per year.

This significant disparity in operating hours, as discussed in Section IV-B, translates into a substantial opportunity for productivity gains and cost savings. Robots offer approximately five times the annual operating capacity of a human worker, which could significantly increase the potential disassembly throughput. While an initial investment in robotics is necessary, the heightened output and reduced labor costs per battery pack could result in considerable long-term economic benefits. Adopting such technologies is poised to enhance annual throughput, thereby improving the overall efficiency of the recycling process for EV LIBs.

VI. CONCLUSION

The process of pack-to-module (P2M) and module-to-cell (M2C) disassembly of the Mitsubishi Outlander PHEV battery pack was explored from a process and design perspective. Based on a tear-down analysis of the disassembly

process, the most labor / cost-intensive parts were identified. To explore difficulties from a process automation perspective, the autonomy of each task was estimated. Based on the proposed automation taxonomy, most P2M disassembly time was readily automatable (57%), with a further 24% partially automatable with human interventions during the task or with prior knowledge of fastener locations or required robot motions.

Analysis of automation of the tasks with the highest level of autonomy yielded an overall ratio of disassembly times before and after robotizing of near unity, whereas it was slightly higher (1.05) for the M2C process. To exemplify the impact of this on the existing disassembly process, an economic analysis of the disassembly process was carried out based on a case study of a human-robot collaborative workstation where a manual two-operator workstation was replaced with a single operator with cobot assistance. We found incorporating robots into the disassembly process based on the assessed currently automatable tasks reduced disassembly cost for the existing P2C process by 43.75%.

Looking forward, future research will focus on exploring various battery types, including a diverse set of battery designs beyond the array-based approach currently considered, to benchmark efficiency factors and facilitate standardized approaches for evaluating automation potential. Exploring business models that enable SMEs and OEMs to leverage economic analyses could expand automation across sectors, increasing revenues and promoting sustainable practices. Another important avenue for future work is optimizing the coordination of multiple robots to improve disassembly line balancing. This optimized multi-robot setup would require the same technical evaluation and modified efficiency factor. Similarly, using the economic analysis framework established in this study, we would quantify the potential economic gains achieved through optimized multi-robot coordination. Overall, this study showcases the significant potential for robotics to improve EV battery disassembly, enhancing economic viability and contributing to a more sustainable circular economy.

APPENDIX

ALIASING AND UNDER-COMMITMENT OF ROBOT RESOURCES

Aliasing effects become significant for small numbers of workers or disassembly workstations, or where human and robot resources are not shared between workstations. In addition, the task feasibility constraints imposed by automation potential of tasks imposes an effective upper bound on autonomy in a practical disassembly setup. Given the allocation of human and robot workers to achieve a given value of autonomy as (15), the mixture of technicians to robots for a given total number of workers N_w can be calculated as:

$$N_e(\alpha) = \left\lceil \frac{1 - \alpha}{1 + \frac{\alpha}{\eta} - \alpha} N_w \right\rceil, \quad (27)$$

$$N_r(\alpha) = \left\lceil \frac{1}{1 + \frac{\eta}{\alpha} - \eta} N_w \right\rceil. \quad (28)$$

for a given number of employees, the number of robots can alternatively be calculated as:

$$N_r(\alpha) = \left\lceil \frac{\alpha}{\eta(1 - \alpha)} N_e \right\rceil \quad (29)$$

given the autonomy upper bound imposed by task feasibility constraints as (14), labelled α_{max} , the effective number of human and robot workers follows as $N_{e,eff} = N_e(\alpha_{eff})$, and similarly for N_r , where

$$\alpha_{eff} = \min(\alpha, \alpha_{max}). \quad (30)$$

$\alpha \geq \alpha_{max}$ implies under-commitment, and therefore, inefficient use of robot resources to fulfil available tasks. The full disassembly profitability can be calculated as the following, considering aliasing effects and the effective maximum autonomy given by the task feasibility constraints:

$$I = (N_{e,eff} + \eta N_{r,eff}) V_B N_B - N_{e,eff} c_e - N_r c_r \quad (31)$$

noting that only the cost of human workers is bounded by α_{max} , while total robot cost is unbounded.

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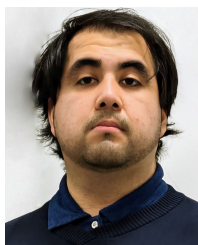
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project, which focuses on robotizing the process of testing, disseminating, and sorting EV batteries.

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